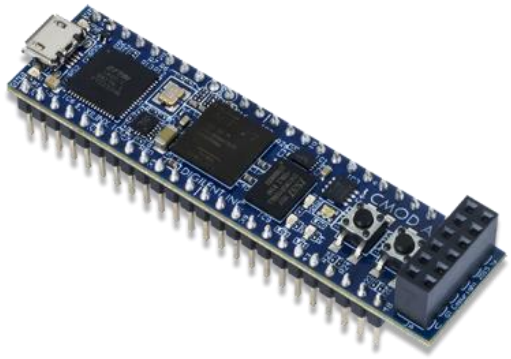
1300 Henley Court
Pullman, WA 99163
509.334.6306
www.digilentinc.com

# Cmod A7 Reference Manual

**Revised October 4, 2019**
**This manual applies to the Cmod A7 Rev. B**

## Overview

The Digilent Cmod A7 is a small, 48-pin DIP form factor board built around a Xilinx Artix-7 FPGA. The board also includes a USB-JTAG programming circuit, USB-UART bridge, clock source, Pmod host connector, SRAM, Quad-SPI Flash, and basic I/O devices. These components make it a formidable, albeit compact, platform for digital logic circuits and MicroBlaze™ embedded soft-core processor designs alike. There are 44 Digital FPGA I/O signals and two FPGA Analog inputs that are routed to 100-mil-spaced through-hole pins so that users can integrate programmable logic design directly into a solderless breadboard circuit. At just 0.7" by 2.75", it can also be loaded in a standard socket and used in embedded systems.



*The Cmod A7.*

- **System Features**
  - 512KB SRAM with an 8-bit bus and 8ns access times
  - 4MB Quad-SPI Flash
  - USB-JTAG Programming Circuitry
  - Powered from USB or external 3.3-5.5V supply connected to DIP pins
- **System Connectivity**
  - USB-UART bridge
- **Interaction and Sensory Devices**
  - 2 LEDs
  - 1 RGB LED
  - 2 Push Buttons
- **Expansion Connectors**
  - 48-pin DIP connector with 44 Digital I/O and 2 Analog inputs (0-3.3V)
  - One Pmod connector with 8 Digital I/O

The Cmod A7 can be purchased with either an Artix-15T or Artix-35T FPGA. These two Cmod A7 product variants are referred to as the Cmod A7-15T and Cmod A7-35T, respectively. When Digilent documentation describes functionality that is common to both of these variants, they are referred to collectively as the "Cmod A7". When describing something that is only common to a specific variant, the variant will be explicitly called out by its name.

The only difference between the Cmod A7-15T and Cmod A7-35T are the capabilities of the FPGA found on the board. They compare as follows:


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| Product Variant | Cmod A7-15T | Cmod A7-35T |
|---|---|---|
| FPGA Part | XC7A15T-1CPG236C | XC7A35T-1CPG236C |
| 1 MSPS On-chip ADC | Yes | Yes |
| Programming options | Quad-SPI Flash/JTAG | Quad-SPI Flash/JTAG |
| Look-up Tables (LUTs) | 10,400 | 20,800 |
| Flip-Flops | 20,800 | 41,600 |
| Block RAM | 112.5 KB | 225 KB |
| Clock Management Tiles | 5 | 5 |

The Cmod A7 is fully compatible with the high-performance Vivado® Design Suite. It is supported under the free WebPACK™ license, so designs can be implemented at no additional cost. This free license includes the ability to create MicroBlaze soft-core processor designs. Design resources, example projects, and tutorials are available for download at the Cmod A7 Resource Center.

# 1 Power Supplies

The Cmod A7 uses a Linear Technologies LTC3569 triple output buck power regulator to generate the needed voltages for the onboard components. It is connected as shown in Fig. 1.1. The characteristics of the outputs are shown in Table 1.1.

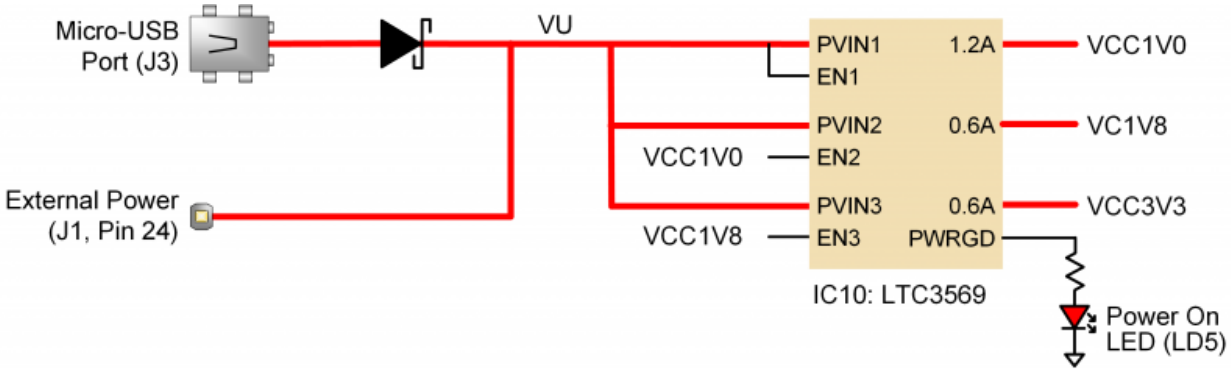



*Figure 1.1. Cmod A7 Power Supply.*

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| Net name | Upstream net name | Power IC Type | Power IC Label | Min/Typ/Max Voltage | Max. Current | Major Devices and Connectors |
|---|---|---|---|---|---|---|
| VCC1V0 | VU | Switching PS | IC10 | 0.95V/1.00V/1.05V | 1.2A | FPGA Core and Block RAM |
| VCC1V8 | VU | Switching PS | IC10 | 1.71V/1.80V/1.89V | 0.6A | FPGA AUX and ADC, USB Core |
| VCC3V3 | VU | Switching PS | IC10 | 3.135V/3.30V/3.465V | 0.6A | SRAM, USB Controller, PMOD, LEDS, Buttons, FPGA USER I/O |

*Table 1.1. Power Rail Characteristics.*

## 1.1 Power Input Options

It is possible to power the Cmod A7 from either USB or from an external power supply attached to pins 24 and 25 of the DIP connector, labeled “VU” and “GND”, respectively. The required characteristics of a power source attached to either of these two options is outlined in Table 1.1.1.

| Connector Type | Connector Label | Schematic net name | Min/Rec/Max Voltage |
|---|---|---|---|
| USB | J3 | USB5V0 | 4.5/5/5.5 |
| DIP | J1 (Pin 24) | VU | 3.32/5/5.5 |

*Table 1.1.1. Power source options.*

The minimum external voltage that can be applied to VU depends on the amount of current drawn from the VCC3V3 rail via the Pmod header. With 0mA drawn from the Pmod header, the minimum external voltage is 3.32V. With 100mA drawn from the Pmod header, the minimum external voltage is 3.38V. With 250mA drawn from the Pmod header, the minimum external voltage is 3.48V.

When the Cmod A7 is being powered by the USB connector, the voltage from the USB device is driven onto the VU pin. This allows users to power an external circuit from the USB host in addition to the Cmod A7. The VU pin is driven via a schottky diode, so a small voltage drop will occur, but it should be assumed the voltage present on VU will be close to 5V.

**Warning: When a USB host is attached to the Micro-USB connector, the VU pin on the DIP connector (pin 24) is driven to the voltage being provided by the USB host (typically between 4.5V-5.5V). If you have a power source attached to the VU pin, you must disconnect it before attaching a USB host, or risk damaging it. This can be particularly dangerous if the power source is a battery.**

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# 2 FPGA Configuration

After power-on, the Artix-7 FPGA must be configured (or programmed) before it can perform any functions. You can configure the FPGA in one of two ways:

1. A PC can use the Digilent USB-JTAG circuitry to program the FPGA any time the power is on.
2. A file stored in the nonvolatile serial (Quad-SPI) flash device can be transferred to the FPGA using the SPI port.

On power-up the Cmod A7 is programmed by the Quad-SPI flash. If the FPGA does not find a valid configuration file in the flash, then it will remain unconfigured until it is programmed over JTAG.

The FPGA configuration data is stored in files called bitstreams that have the .bit file extension. The Vivado software from Xilinx can create bitstreams from VHDL or Verilog® based source files.

Bitstreams are stored in volatile memory cells within the FPGA. This data defines the FPGA's logic functions and circuit connections, and it remains valid until it is erased by removing board power or by writing a new configuration file using the JTAG port.

Both the Artix-7 35T and 15T bitstreams are typically 17,536,096 bits. The time it takes to program the Cmod A7 can be decreased by compressing the bitstream before programming, and then allowing the FPGA to decompress the bitstream itself during configuration. Depending on design complexity, compression ratios of 10x can be achieved. Bitstream compression can be enabled within the Vivado tools to occur during generation. For instructions on how to do this, consult the Xilinx documentation for Vivado.

After being successfully programmed, the FPGA will cause the "DONE" LED to illuminate.

The following sections provide greater detail about programming the Cmod A7 using the different methods available.

## 2.1 JTAG Configuration

The Xilinx tools typically communicate with FPGAs using the Test Access Port and Boundary-Scan Architecture, commonly referred to as JTAG. During JTAG programming, a .bit file is transferred from the PC to the FPGA using the onboard Digilent USB-JTAG circuitry (micro USB port). You can perform JTAG programming any time after the Cmod A7 has been powered on. If the FPGA is already configured, then the existing configuration is overwritten with the bitstream being transmitted over JTAG.

Programming the Cmod A7 with an uncompressed bitstream using the on-board USB-JTAG circuitry usually takes around 6 seconds. JTAG programming can be done using the hardware manager in Vivado.

**Warning: When a USB host is attached to the micro USB connector, the VU pin on the DIP connector (pin 24) is driven to the voltage being provided by the USB host (typically between 4.5V-5.5V). If you have a power source attached to the VU pin, you must disconnect it before attaching a USB host, or risk damaging it. This can be particularly dangerous if the power source is a battery.**

## 2.2 Quad-SPI Configuration

Since the FPGA's memory on the Cmod A7 is volatile, it relies on the Quad-SPI flash memory to store the configuration between power cycles. This configuration mode is referred to in Xilinx documentation as Master SPI. The blank FPGA takes the role of master and reads the configuration file out of the flash device upon power-up. To

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that effect, a configuration file needs to be downloaded first to the flash. When programming a nonvolatile flash device, a bitstream file is transferred to the flash in a two-step process. First, the FPGA is programmed with a circuit that can program flash devices, and then data is transferred to the flash device via the FPGA circuit (this complexity is hidden from the user by the Xilinx tools). This is called indirect programming. After the flash device has been programmed, it can automatically configure the FPGA at a subsequent power-on event. Programming files stored in the flash device will remain until they are overwritten, regardless of power-cycle events.

Programming the flash can take as long as four to five minutes, which is mostly due to the lengthy erase process inherent to the memory technology. Once written, however, FPGA configuration can be very fast—less than a second. Bitstream compression, SPI bus width, and configuration rate are factors controlled by the Xilinx tools that can affect configuration speed. The Cmod A7 supports x1, x2, and x4 bus widths and data rates of up to 50 MHz for Quad-SPI programming.

Quad-SPI programming can be done using the hardware manager in Vivado.

# 3 SRAM

The Cmod A7 includes 512 KB of Static Random-Access Memory (SRAM). This memory has a standard, easy-to-use parallel interface with 19 address signals, 8 bi-directional data signals, and 3 control signals. The part used is the ISSI IS61WV5128BLL-10BLI. The interface is fully documented in ISSI's datasheet for the part.

The power supply used to power the SRAM provides 3.3V with greater than 5% accuracy. This allows the IS61WV5128BLL-10BLI part to be written to and read from with 8 ns access times. This allows for a theoretical maximum data bandwidth of 125,000,000 B/s.

The onboard SRAM is very useful for MicroBlaze designs that would like to provide the processor with additional memory. To do this, the SRAM can be controlled with Xilinx External Memory Controller IP core. The Cmod A7 User demo available on the Cmod A7 resource center demonstrates how to do this.

For pin-mappings between the FPGA and SRAM, please see the CmodA7 Master XDC available on the Cmod A7 Resource Center.

# 4 Quad-SPI Flash

FPGA configuration files can be written to the Quad-SPI Flash (Micron part number N25Q032A) and the FPGA will automatically configure itself with it after power on (as described in section 2.2). An Artix-7 35T or 15T configuration file requires 17,536,096 bits of memory, leaving about half of the flash device (just over 2MB) available for user data. A common use for this extra memory is to store MicroBlaze programs too big to fit in the on-chip Block memory (typically 128 KB for the 35T and 64KB for the 15T). These programs are then loaded and executed using a smaller bootloader program that can fit in the block memory. It is possible to automatically generate this bootloader, roll it into a single file (called an .mcs file) that also contains the bitstream and your custom MicroBlaze application, and program this file into SPI Flash using Xilinx SDK and Vivado. Xilinx Answer Record 63605 explains how to do this.

The contents of the memory can be manipulated by issuing certain commands on the SPI bus. The implementation of this protocol is outside the scope of this document. All signals in the SPI bus are general-purpose user I/O pins after FPGA configuration. On other boards, SCK is an exception because it remains a dedicated pin even after

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configuration, however, on the Cmod A7 the SCK signal is routed to an additional general purpose pin that can be accessed after configuration (pin E19). This allows access to this pin without having to instantiate the special FPGA primitive called STARTUPE2.

Xilinx's AXI Quad SPI core can be used to read/write the flash in a MicroBlaze design. Refer to Xilinx's product guide for this core to learn more about using it, or to Micron's datasheet for the flash device to learn how to implement a custom controller.

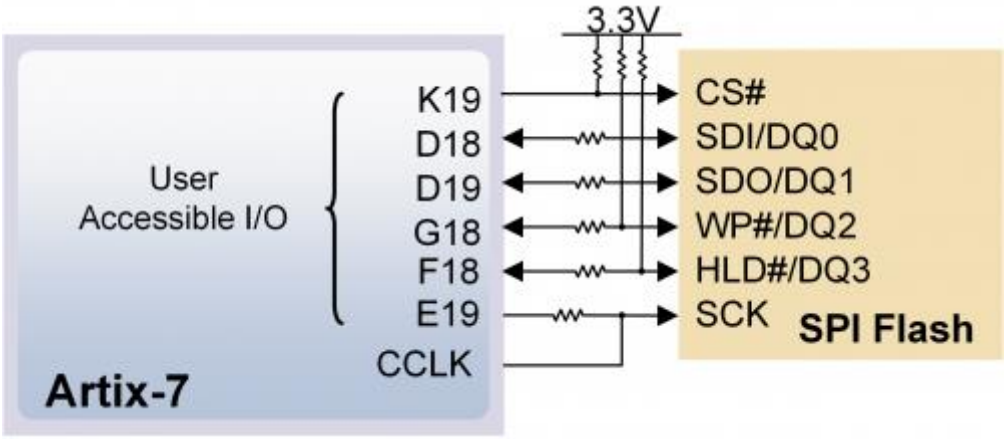



*Figure 4.1. Cmod A7 Quad-SPI Flash.*

# 5 Oscillators/Clocks

The Cmod A7 board includes a single 12 MHz clock input connected to pin L17 (L17 is a MRCC input on bank 14). The input clock can drive MMCMs to generate clocks of various frequencies and with known phase relationships that may be needed throughout a design. The 12 MHz input clock cannot directly drive a PLL because they have a minimum input frequency of 19 MHz. Some rules restrict which MMCMs may be driven by the 12 MHz input clock. For a full description of these rules and of the capabilities of the Artix-7 clocking resources, refer to the "7 Series FPGAs Clocking Resources User Guide" available from Xilinx.

Xilinx offers the Clocking Wizard IP core to help users generate the different clocks required for a specific design. This wizard will properly instantiate the needed MMCMs and PLLs based on the desired frequencies and phase relationships specified by the user. The wizard will then output an easy-to-use wrapper component around these clocking resources that can be inserted into the user's design. The clocking wizard can be accessed from within the Project Navigator or Core Generator tools.

# 6 USB-UART Bridge

The Cmod A7 includes an FTDI FT2232HQ USB-UART bridge (attached to the Micro-USB connector) that allows users to use PC applications to communicate with the board using standard Windows COM port commands. Free USB-COM port drivers, available from www.ftdichip.com under the "Virtual Com Port" or VCP heading, convert USB packets to UART/serial port data. Serial port data is exchanged with the FPGA using a two-wire serial port (TXD/RXD). After the drivers are installed, I/O commands can be used from the PC directed to the COM port to produce serial data traffic on the J17 and J18 FPGA pins.

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An on-board status LED provides visual feedback on traffic flowing through the port: the LED next to DIP pin 25 will blink in response to TX or RX traffic.

The FT2232HQ is also used as the controller for the Digilent USB-JTAG circuitry, but the USB-UART and USB-JTAG functions behave entirely independent of one another. Programmers interested in using the UART functionality of the FT2232 within their design do not need to worry about the JTAG circuitry interfering with the UART data transfers, and vice-versa. The combination of these two features into a single device allows the Cmod A7 to be programmed, communicated with via UART, and powered from a computer attached with a single Micro-USB cable.

**Warning: When a USB host is attached to the Micro-USB connector, the VU pin on the DIP connector (pin 24) is driven to the voltage being provided by the USB host (typically between 4.5V-5.5V). If you have a power source attached to the VU pin, you must disconnect it before attaching a USB host, or risk damaging it. This can be particularly dangerous if the power source is a battery.**

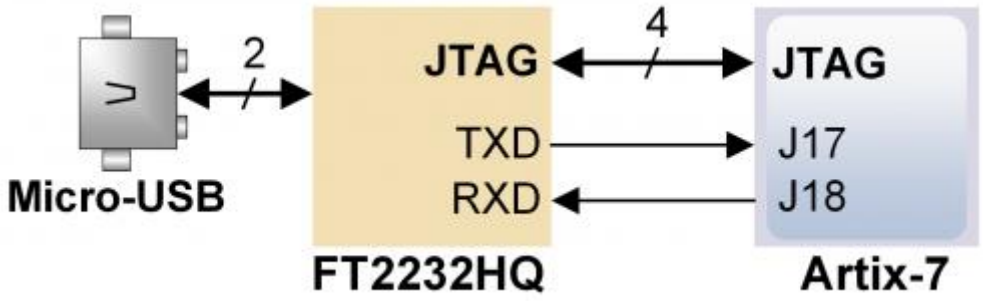



*Figure 5.1. Cmod A7 Quad-SPI Flash.*

# 7 Basic I/O

The Cmod A7 includes a tri-color LED, two push buttons, and two individual LEDs. The push buttons are connected to the FPGA via series resistors to prevent damage from inadvertent short circuits (a short circuit could occur if an FPGA pin assigned to a push button was inadvertently defined as an output). The two push buttons are momentary switches that normally generate a low output when they are at rest, and a high output only when they are pressed.

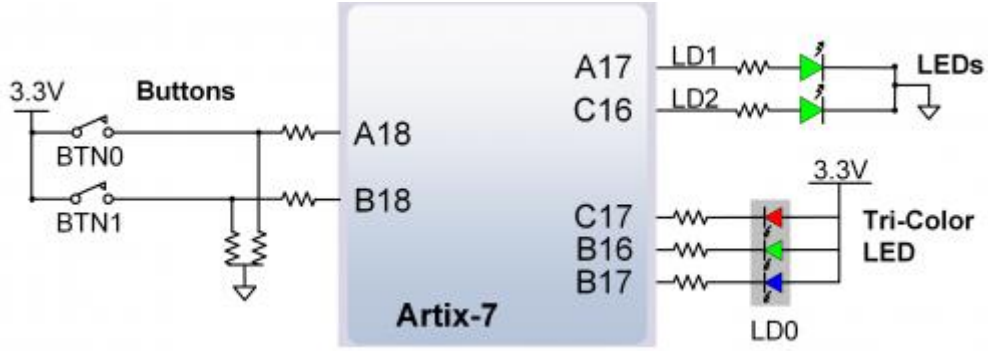



*Figure 6.1. Cmod A7 Basic I/O.*

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The two individual high-efficiency LEDs are anode-connected to the FPGA via 330-ohm resistors, so they will turn on when a logic high voltage is applied to their respective I/O pin. Additional LEDs that are not user-accessible indicate power-on, FPGA programming status, and USB-UART traffic.

## 7.1 Tri-color LED

The Cmod A7 contains a tri-color LED. The tri-color LED has three smaller internal LEDs: one red, one blue, and one green. The anodes of these LEDs are connected to the 3.3V rail, and the cathodes are connected to FPGA signals. Driving one of the RGB LED signals connected to the FPGA low will cause current to flow through the corresponding LED, illuminating it. The tri-color LED will emit a color dependent on the combination of internal LEDs that are currently being illuminated. For example, if the red and blue signals are driven low and green is driven high, the tri-color LED will emit a purple color.

**Note: Digilent strongly recommends the use of pulse-width modulation (PWM) when driving the tri-color LED. Driving any of the inputs to a steady logic '1' will result in the LED being illuminated at an uncomfortably bright level. You can avoid this by ensuring that none of the tri-color signals are driven with more than 50% duty cycle. Using PWM also greatly expands the potential color palette of the tri-color LED. Individually adjusting the duty cycle of each color between 5-% and 0% causes the different colors to be illuminated at different intensities, allowing virtually any color to be displayed.**

# 8 DIP Connector

The Cmod A7 has a 48-pin DIP connector for connecting to breadboards and custom fixtures. The pins have 100-mil spacing, and the entire module is 0.7 inches by 2.75 inches. Of the 48-pins, 44 are connected directly to FPGA Digital I/Os, two are voltage-divided and connected to FPGA analog inputs, and two are connected to power pins. The pin numbering of the DIP connector is shown in Fig. 8.1.

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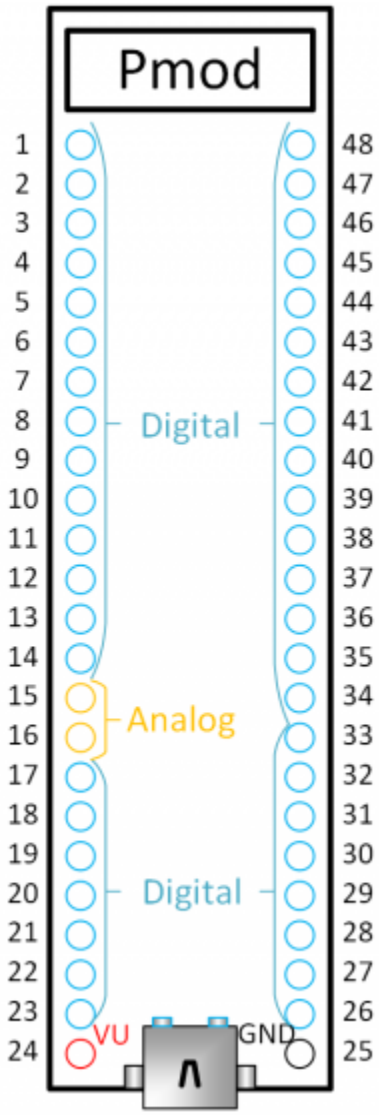



*Figure 8.1. DIP pin diagram.*

VU (pin 24) can be used to power the Cmod A7 when it is not connected to a USB host. When the device is connected to a USB host, VU is driven by the 5V rail of the USB connector and can be used to power devices external to the Cmod A7. See the “Power” section of this manual for more information on powering the Cmod A7.

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## 8.1 DIP Digital I/O

The pins connected directly to the FPGA can be used as general purpose inputs or outputs. There are no series resistors between the FPGA and the DIP pins, so care should be taken not to cause shorts or connect them to voltages greater than allowed by the FPGA. The absolute maximum voltages for these pins are outlined in Table 8.1.

| | Absolute Minimum Voltage | Recommended Minimum Operating Voltage | Recommended Maximum Operating Voltage | Absolute Maximum Voltage |
|---|---|---|---|---|
| Powered from USB or VU | -0.4 V | -0.2 V | 3.4 V | 3.75 V |
| Unpowered | -0.4 V | N/A | N/A | 0.55 V |

*Table 8.1. DIP pin voltage ratings.*

For more information on the electrical characteristics of the pins connected to the FPGA, see the Artix-7 datasheet from Xilinx.

## 8.2 Analog Inputs

Pins 15 and 16 of the DIP connector are used as analog inputs to the XADC module of the FPGA. The FPGA expects that the inputs range from 0-1V, so we use an external circuit to scale down the input voltage from 3.3V. This circuit is shown in Fig. 8.2.1. This circuit allows the XACD module to accurately measure any voltage between 0V and 3.3V (relative to GND on pin 25) that is applied to either of these pins.

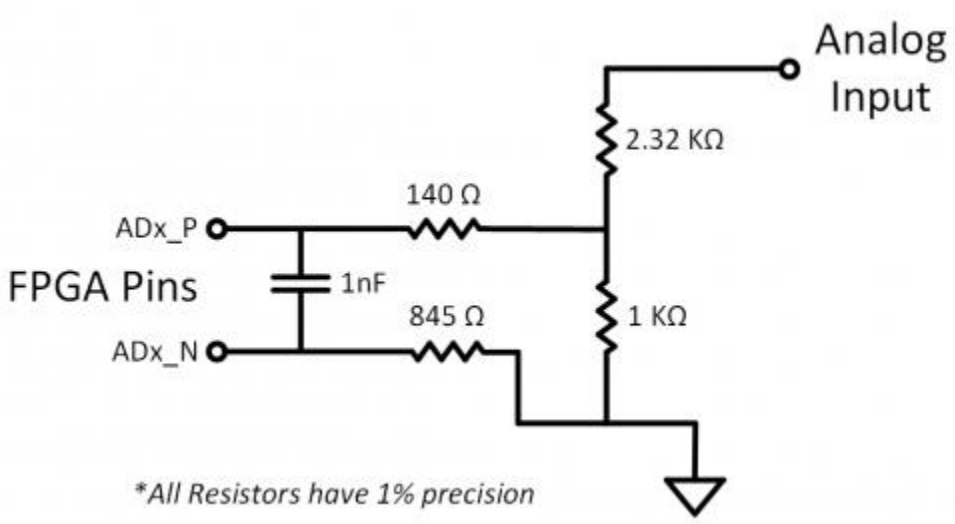



*Figure 8.2.1. Analog input circuit.*

The XADC core within the Artix-7 is a dual-channel 12-bit analog-to-digital converter capable of operating at 1 MSPS. Either channel can be driven by any of the two auxiliary analog inputs connected to the DIP pins. The XADC core is controlled and accessed from a user design via the Dynamic Reconfiguration Port (DRP). The DRP also provides access to voltage monitors that are present on each of the FPGA's power rails, and a temperature sensor that is internal to the FPGA. For more information on using the XADC core, refer to the Xilinx document titled 7 Series FPGAs and Zynq-7000 All Programmable SoC XADC Dual 12-bit 1 MSPS Analog-to-Digital Converter. A demo that uses the XADC core is available on the Cmod A7 resource center.

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# 9 Pmod Connector

Pmod connecters are 2x6, right-angle, 100-mil spaced female connectors that mate with standard 2x6 pin headers. Each 12-pin Pmod connector provides two 3.3V VCC signals (pins 6 and 12), two ground signals (pins 5 and 11), and eight logic signals, as shown in Fig. 9.1. The VCC and Ground pins can deliver up to 1A of current, but care must be taken not to exceed any of the power budgets of the onboard regulators of the external power supply (these are described in the “Power Supplies” section).

Digilent produces a large collection of Pmod accessory boards that can attach to the Pmod expansion connectors to add ready-made functions like A/Ds, D/As, motor drivers, sensors, and other functions. See www.digilentinc.com for more information.

The Cmod A7 has one standard-type Pmod connector, and the FPGA pin connections for it are described in Table 9.1. Standard Pmod connectors are connected to the FPGA via 200-ohm series resistors. The series resistor prevent short circuits that can occur if the user accidentally drives a signal that is supposed to be used as an input. The downside to this added protection is that these resistors can limit the maximum switching speed of the data signals.

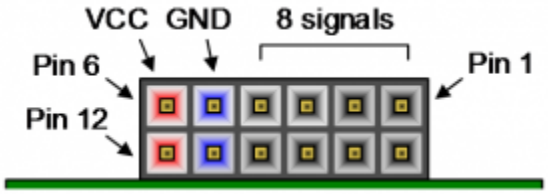
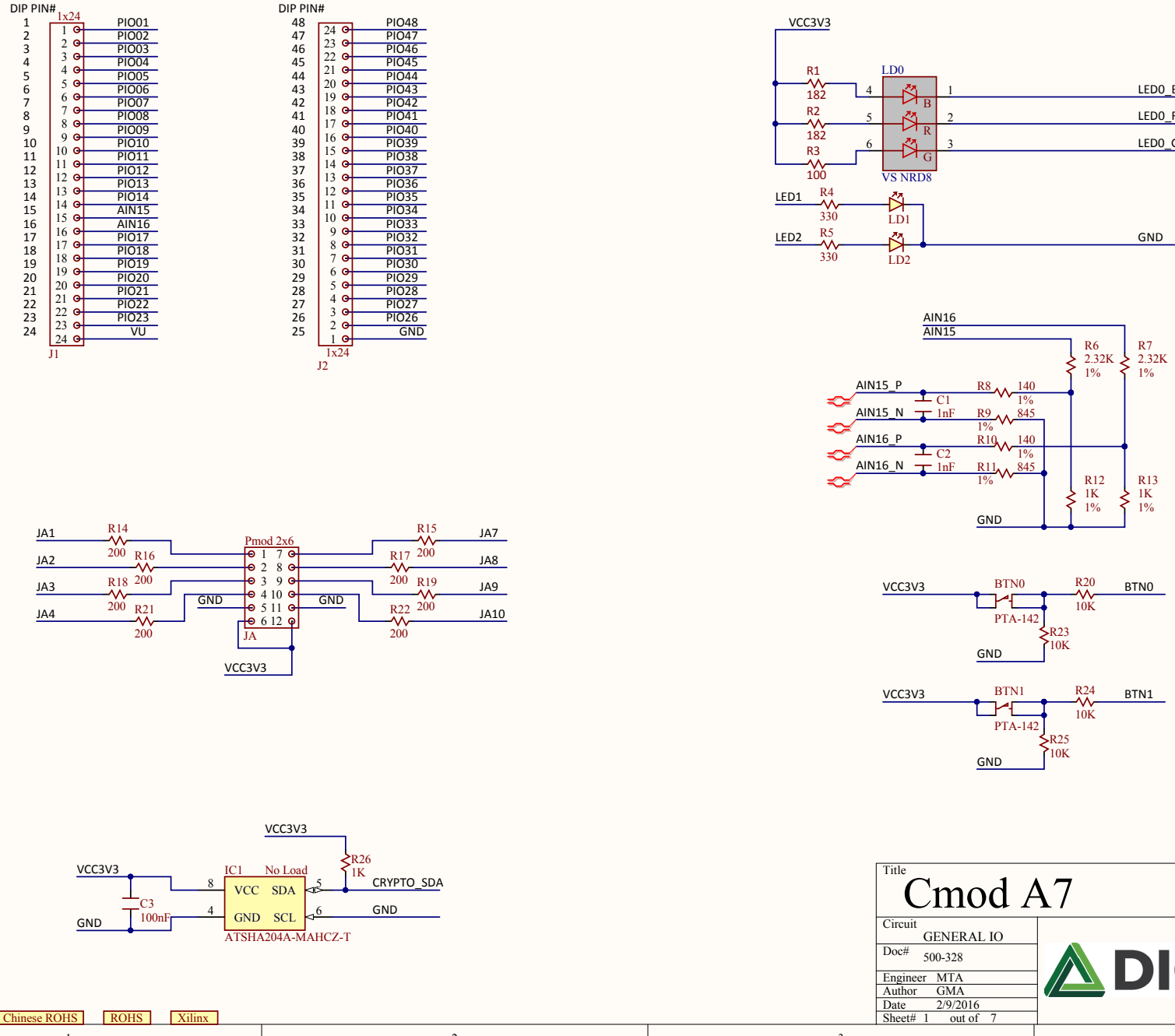


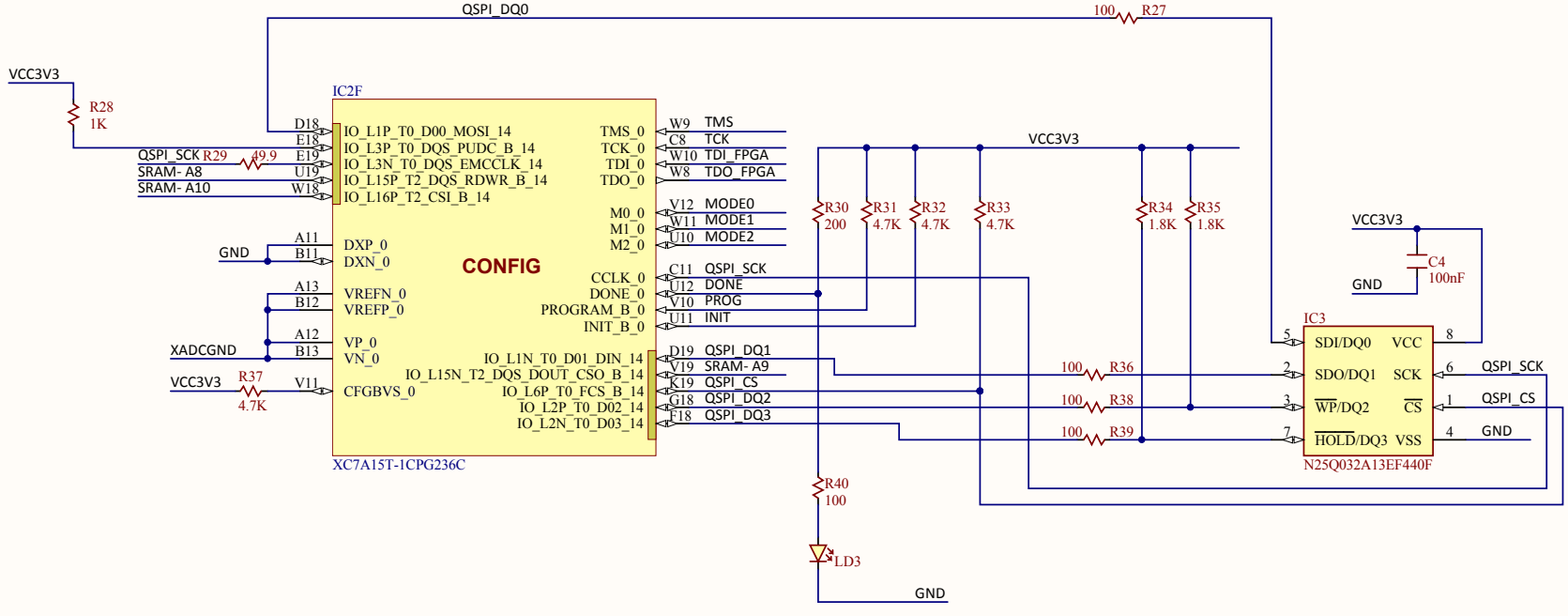

*Figure 9.1. Pmod diagram.*

| | Pmod JA |
|---|---|
| **Pmod Type** | **Standard** |
| Pin 1 | G17 |
| Pin 2 | G19 |
| Pin 3 | N18 |
| Pin 4 | L18 |
| Pin 7 | H17 |
| Pin 8 | H19 |
| Pin 9 | J19 |
| Pin 10 | K18 |

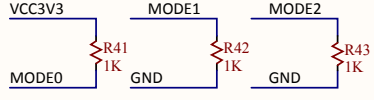
*Table 9.1. Cmod A7 Pmod pinout.*

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DIP PIN#
1x24
PIO01
PIO02
PIO03
PIO04
PIO05
PIO06
PIO07
PIO08
PIO09
PIO10
PIO11
PIO12
PIO13
PIO14
AIN15
AIN16
PIO17
PIO18
PIO19
PIO20
PIO21
PIO22
PIO23
VU
J1
DIP PIN#
PIO48
PIO47
PIO46
PIO45
PIO44
PIO43
PIO42
PIO41
PIO40
PIO39
PIO38
PIO37
PIO36
PIO35
PIO34
PIO33
PIO32
PIO31
PIO30
PIO29
PIO28
PIO27
PIO26
GND
1x24
J2
VCC3V3
R1
182
LD0
B
R
G
VS NRD8
LED0_B
LED0_R
LED0_G
R2
182
R3
100
LED1
R4
330
LD1
LED2
R5
330
LD2
GND
AIN16
AIN15
R6
2.32K
1%
R7
2.32K
1%
AIN15_P
AIN15_N
AIN16_P
AIN16_N
C1
1nF
C2
1nF
R8
140
1%
R9
845
1%
R10
140
1%
R11
845
1%
R12
1K
1%
R13
1K
1%
GND
JA1
JA2
JA3
JA4
R14
200
R16
200
R18
200
R21
200
Pmod 2x6
JA
R15
200
R17
200
R19
200
R22
200
JA7
JA8
JA9
JA10
GND
GND
VCC3V3
VCC3V3
BTN0
R20
10K
BTN0
PTA-142
R23
10K
GND
VCC3V3
BTN1
R24
10K
BTN1
PTA-142
R25
10K
GND
VCC3V3
R26
1K
VCC3V3
IC1
No Load
VCC
SDA
GND
SCL
CRYPTO_SDA
GND
C3
100nF
GND
ATSHA204A-MAHCZ-T
CE
Chinese ROHS
ROHS
Xilinx
Title
Cmod A7
Rev
B.1
Copyright 2015
Circuit
GENERAL IO
Doc#
500-328
Engineer MTA
Author GMA
Date 2/9/2016
Sheet# 1 out of 7
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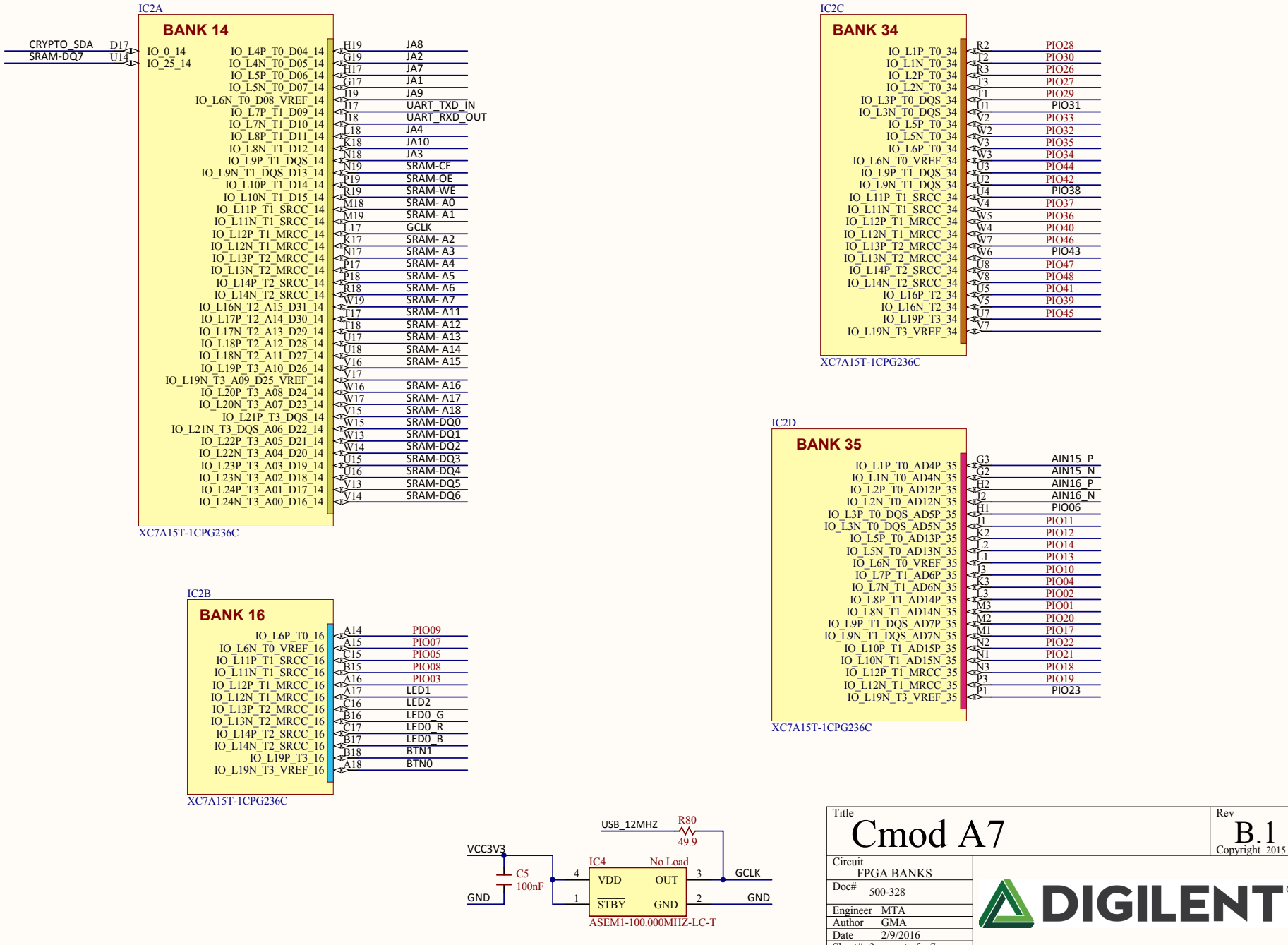


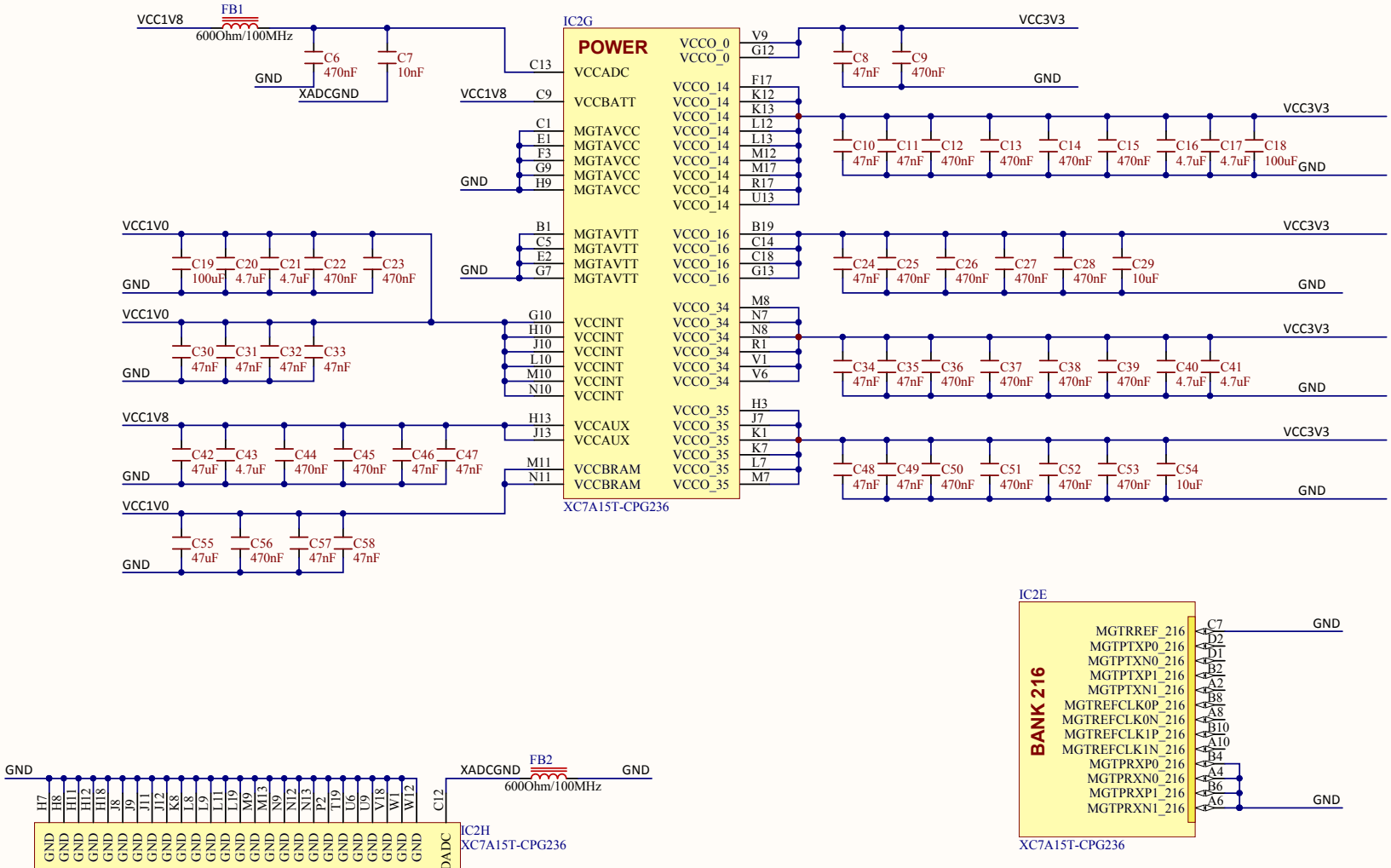

Defaults to SPI Programming




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|---|---|
| Title | Cmod A7 |
| Rev | B.1 |
| Copyright | 2015 |
| Circuit | CONFIG, SPI FLASH |
| Doc# | 500-328 |
| Engineer | MTA |
| Author | GMA |
| Date | 2/9/2016 |
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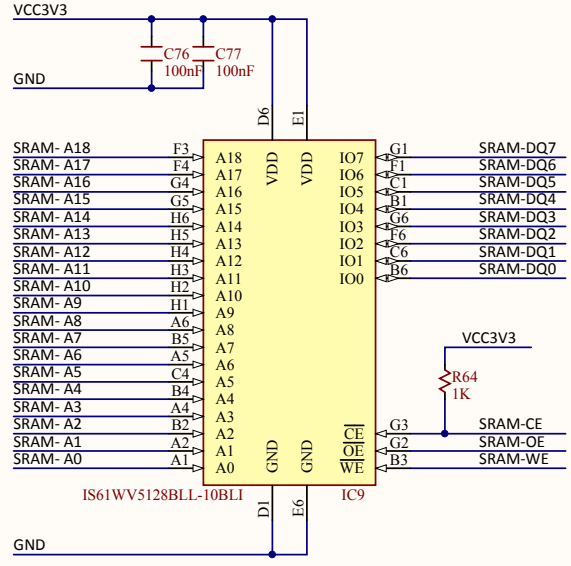

IC2A
BANK 14
CRYPTO_SDA D17
SRAM-DQ7 U14
IO_0_14
IO_25_14
IO_L4P_T0_D04_14 H19 JA8
IO_L4N_T0_D05_14 G19 JA2
IO_L5P_T0_D06_14 H17 JA7
IO_L5N_T0_D07_14 G17 JA1
IO_L6N_T0_D08_VREF_14 J19 JA9
IO_L7P_T1_D09_14 J17 UART_TXD_IN
IO_L7N_T1_D10_14 J18 UART_RXD_OUT
IO_L8P_T1_D11_14 L18 JA4
IO_L8N_T1_D12_14 K18 JA10
IO_L9P_T1_DQS_14 N18 JA3
IO_L9N_T1_DQS_D13_14 N19 SRAM-CE
IO_L10P_T1_D14_14 P19 SRAM-OE
IO_L10N_T1_D15_14 R19 SRAM-WE
IO_L11P_T1_SRCC_14 M18 SRAM- A0
IO_L11N_T1_SRCC_14 M19 SRAM- A1
IO_L12P_T1_MRCC_14 L17 GCLK
IO_L12N_T1_MRCC_14 K17 SRAM- A2
IO_L13P_T2_MRCC_14 N17 SRAM- A3
IO_L13N_T2_MRCC_14 P17 SRAM- A4
IO_L14P_T2_SRCC_14 P18 SRAM- A5
IO_L14N_T2_SRCC_14 R18 SRAM- A6
IO_L16N_T2_A15_D31_14 W19 SRAM- A7
IO_L17P_T2_A14_D30_14 T17 SRAM- A11
IO_L17N_T2_A13_D29_14 T18 SRAM- A12
IO_L18P_T2_A12_D28_14 U17 SRAM- A13
IO_L18N_T2_A11_D27_14 U18 SRAM- A14
IO_L19P_T3_A10_D26_14 V16 SRAM- A15
IO_L19N_T3_A09_D25_VREF_14 V17
IO_L20P_T3_A08_D24_14 W16 SRAM- A16
IO_L20N_T3_A07_D23_14 W17 SRAM- A17
IO_L21P_T3_DQS_14 V15 SRAM- A18
IO_L21N_T3_DQS_A06_D22_14 W15 SRAM-DQ0
IO_L22P_T3_A05_D21_14 W13 SRAM-DQ1
IO_L22N_T3_A04_D20_14 W14 SRAM-DQ2
IO_L23P_T3_A03_D19_14 U15 SRAM-DQ3
IO_L23N_T3_A02_D18_14 U16 SRAM-DQ4
IO_L24P_T3_A01_D17_14 V13 SRAM-DQ5
IO_L24N_T3_A00_D16_14 V14 SRAM-DQ6
XC7A15T-1CPG236C
IC2B
BANK 16
IO_L6P_T0_16 A14 PIO09
IO_L6N_T0_VREF_16 A15 PIO07
IO_L11P_T1_SRCC_16 C15 PIO05
IO_L11N_T1_SRCC_16 B15 PIO08
IO_L12P_T1_MRCC_16 A16 PIO03
IO_L12N_T1_MRCC_16 A17 LED1
IO_L13P_T2_MRCC_16 C16 LED2
IO_L13N_T2_MRCC_16 B16 LED0_G
IO_L14P_T2_SRCC_16 C17 LED0_R
IO_L14N_T2_SRCC_16 B17 LED0_B
IO_L19P_T3_16 B18 BTN1
IO_L19N_T3_VREF_16 A18 BTN0
XC7A15T-1CPG236C
IC2C
BANK 34
IO_L1P_T0_34 R2 PIO28
IO_L1N_T0_34 T2 PIO30
IO_L2P_T0_34 R3 PIO26
IO_L2N_T0_34 T3 PIO27
IO_L3P_T0_DQS_34 T1 PIO29
IO_L3N_T0_DQS_34 U1 PIO31
IO_L5P_T0_34 V2 PIO33
IO_L5N_T0_34 W2 PIO32
IO_L6P_T0_34 V3 PIO35
IO_L6N_T0_VREF_34 W3 PIO34
IO_L9P_T1_DQS_34 U3 PIO44
IO_L9N_T1_DQS_34 U2 PIO42
IO_L11P_T1_SRCC_34 U4 PIO38
IO_L11N_T1_SRCC_34 V4 PIO37
IO_L12P_T1_MRCC_34 W5 PIO36
IO_L12N_T1_MRCC_34 W4 PIO40
IO_L13P_T2_MRCC_34 W7 PIO46
IO_L13N_T2_MRCC_34 W6 PIO43
IO_L14P_T2_SRCC_34 U8 PIO47
IO_L14N_T2_SRCC_34 V8 PIO48
IO_L16P_T2_34 U5 PIO41
IO_L16N_T2_34 V5 PIO39
IO_L19P_T3_34 U7 PIO45
IO_L19N_T3_VREF_34 V7
XC7A15T-1CPG236C
IC2D
BANK 35
IO_L1P_T0_AD4P_35 G3 AIN15_P
IO_L1N_T0_AD4N_35 G2 AIN15_N
IO_L2P_T0_AD12P_35 H2 AIN16_P
IO_L2N_T0_AD12N_35 J2 AIN16_N
IO_L3P_T0_DQS_AD5P_35 H1 PIO06
IO_L3N_T0_DQS_AD5N_35 J1 PIO11
IO_L5P_T0_AD13P_35 K2 PIO12
IO_L5N_T0_AD13N_35 L2 PIO14
IO_L6N_T0_VREF_35 L1 PIO13
IO_L7P_T1_AD6P_35 J3 PIO10
IO_L7N_T1_AD6N_35 K3 PIO04
IO_L8P_T1_AD14P_35 L3 PIO02
IO_L8N_T1_AD14N_35 M3 PIO01
IO_L9P_T1_DQS_AD7P_35 M2 PIO20
IO_L9N_T1_DQS_AD7N_35 M1 PIO17
IO_L10P_T1_AD15P_35 N2 PIO22
IO_L10N_T1_AD15N_35 N1 PIO21
IO_L12P_T1_MRCC_35 N3 PIO18
IO_L12N_T1_MRCC_35 P3 PIO19
IO_L19N_T3_VREF_35 P1 PIO23
XC7A15T-1CPG236C
USB_12MHZ
R80
49.9
VCC3V3
C5
100nF
GND
IC4
No Load
VDD
OUT
STBY
GND
GCLK
GND
ASEM1-100.000MHZ-LC-T
Title
Cmod A7
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FPGA BANKS
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500-328
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Author GMA
Date 2/9/2016
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| Doc# | 500-328 |
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| Author | GMA |
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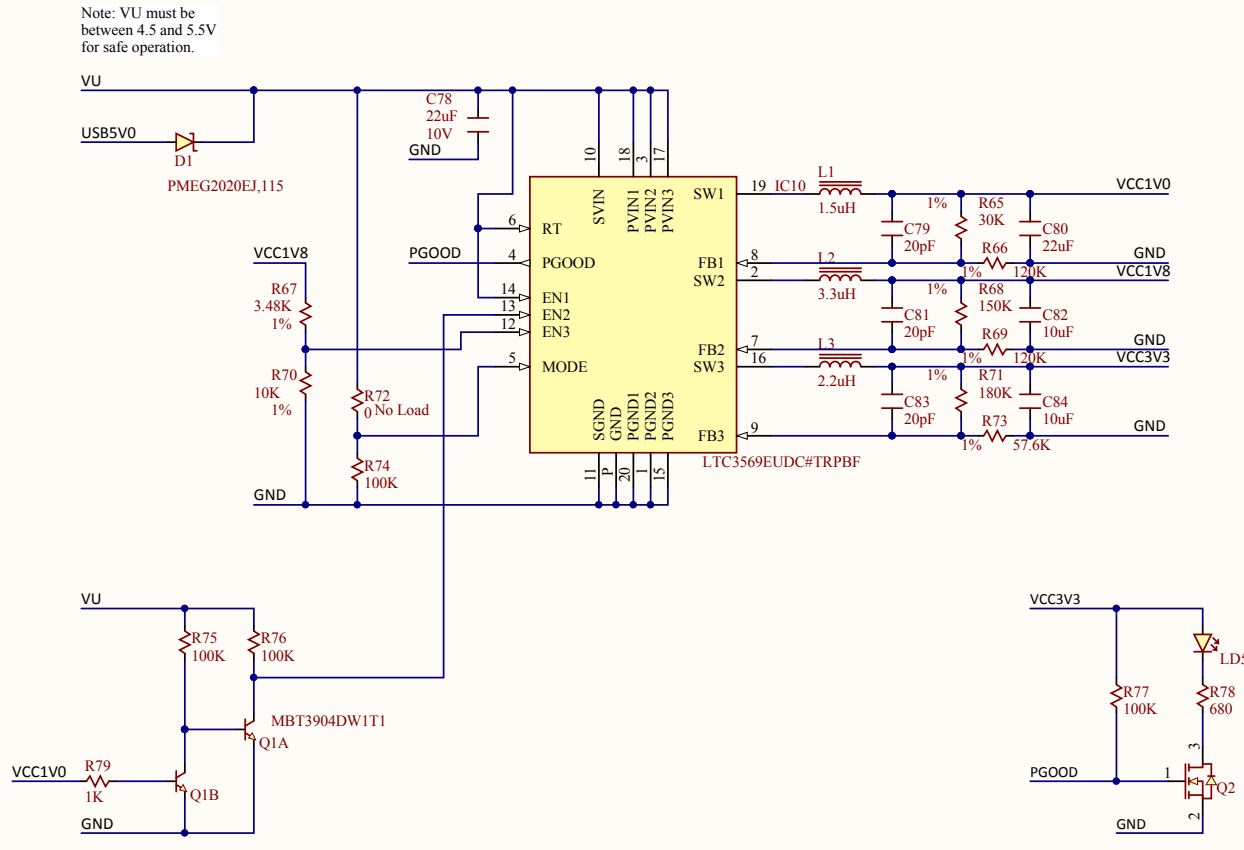
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